CS1380 : 2026 : M6 — Cloud Deployment

Distributed Search Engine

A progressively-built, MapReduce-powered search engine deployed on AWS EC2 — from single-machine crawling to fully distributed indexing and retrieval.

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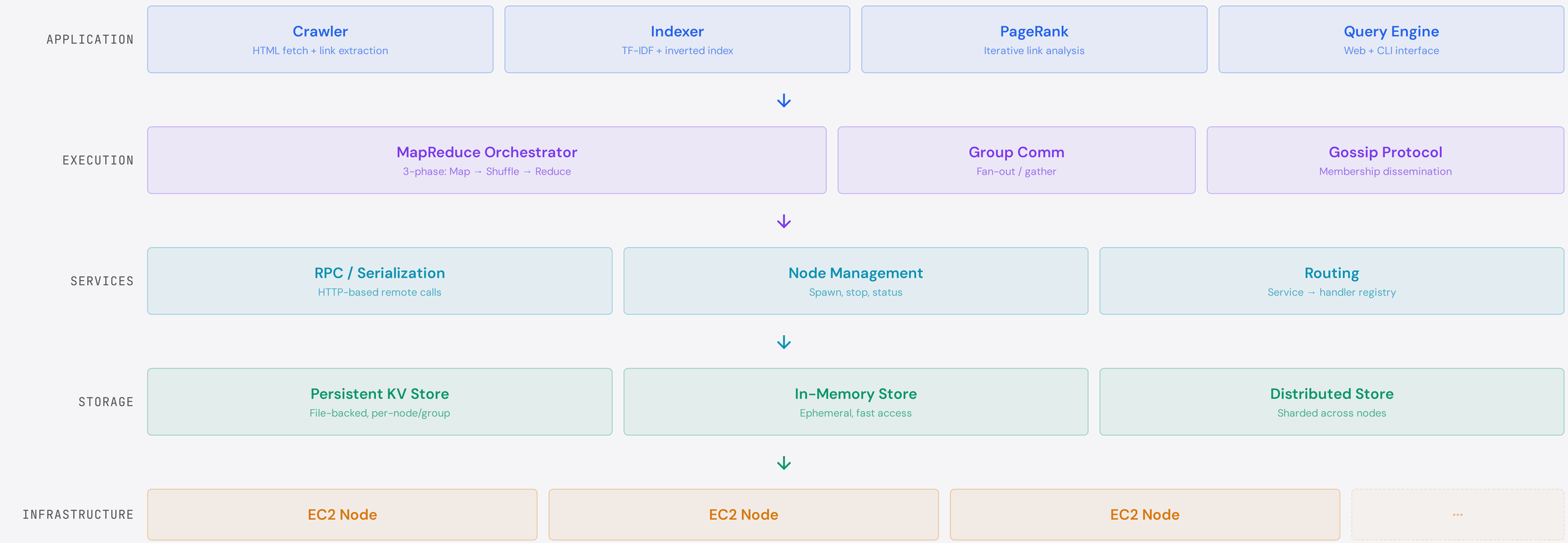
01 — SYSTEM OVERVIEW

End-to-End Architecture

The system is structured in layers: application-level search components communicate through a group-based distributed services framework, backed by persistent and in-memory storage, all running on AWS EC2 nodes.



02 — PROGRESSIVE BUILD

Milestone Progression

M0	**Single-Machine Search** Crawler, indexer, query engine — no distribution	279 LoC
M1	**Serialization** Cross-network object transmission with function support	185 LoC
M2	**Communication** HTTP-based RPC, node spawning, remote invocation	593 LoC
M3	**Groups & Gossip** Group membership, gossip protocol, fan-out/gather	250 LoC
M4	**Distributed Storage** Persistent & in-memory KV stores, sharding by group	1,649 LoC
M5	**MapReduce** Distributed execution with Map → Shuffle → Reduce pipeline	1,899 LoC

~14K Total Lines of Code
59 Hours of Development
5 Milestones Integrated

03 — EXECUTION MODEL

MapReduce Pipeline

The coordinator registers ephemeral services across all group nodes, serializes mapper/reducer functions, and orchestrates three synchronized phases:

MAP — Each node processes assigned keys in parallel → **SHUFFLE** — Consistent hashing redistributes data → **REDUCE** — Nodes aggregate and emit results

Synchronization: Coordinator tracks completion per phase via `notify` callbacks — advances only when all nodes confirm. Functions are serialized and shipped to remote nodes for local execution.

04 — PERFORMANCE

Key Benchmarks

14,250 Requests/sec (parallel RPC) — Local, M2
1,393 Store ops/sec — Local, M4
111.6 Docs/sec (MapReduce) — Local, M5
44.8ms MapReduce latency/run — Local, M5

05 — LOCAL VS. AWS EC2

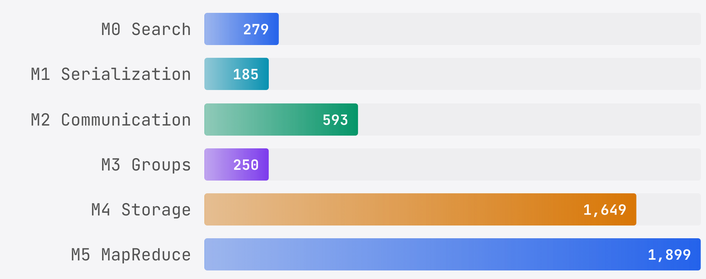
Latency Comparison

OPERATION	LOCAL (APPLE M2)	AWS EC2
Serialization (complex)	44.28 µs	61.83 µs
RPC round-trip	~0.07 ms	~0.36 ms
Node spawn	97.99 ms	146.49 ms
KV Store put	2.12 ms	1.29 ms
KV Store get	0.72 ms	0.89 ms

AWS latency is comparable to local for I/O-bound operations (KV store) but shows network overhead for RPC-heavy workloads. Storage ops benefit from EC2's optimized disk I/O.

06 — THROUGHPUT BY MILESTONE

Lines of Code Contributed



07 — COMPONENTS

Module Breakdown

MODULE	ROLE	LOC
local/node.js	HTTP server, request routing	~200
local/store.js	File-backed persistent KV	~160
local/comm.js	RPC client, error handling	~120
all/mr.js	MapReduce orchestrator	~250
all/comm.js	Fan-out/gather broadcast	~150
util/serialization.js	Object/function serializer	185
non-dist/merge.js	Inverted index merger	171

08 — CHALLENGES & SOLUTIONS

Key Technical Decisions

Serializing Functions Across the Wire

MapReduce requires shipping mapper/reducer functions to remote nodes. Built a recursive serializer handling closures, special values (NaN, Infinity), nested objects, and Date/Error types.

MapReduce Phase Synchronization

Coordinator must wait for all nodes to complete each phase before advancing. Implemented a notify-based callback barrier with ephemeral service registration/deregistration per job.

Consistent Hash-Based Shuffle

During shuffle, all nodes must agree on data routing. Used deterministic consistent hashing on keys to ensure symmetric placement across the cluster.

Group-Scoped Service Isolation

Every service is scoped to a named group, enabling multiple independent clusters (e.g., crawler group, indexer group) on the same physical nodes.

09 — STACK & ENVIRONMENT

Technology

`Node.js` `JavaScript ES6+` `AWS EC2` `HTTP RPC` `Jest` `natural.js` `jsdom` `html-to-text` `yargs`

Dev environment: Apple M2, 16 GB RAM, 8 cores (local). Ubuntu on Amazon EC2 (deployed). All components tested both locally and on multi-node AWS clusters.

Testing: Comprehensive Jest suite with 4 tiers — unit tests, scenario tests (NCDC, TF-IDF, string matching), extra credit, and instructor-provided integration tests.